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REVIEW 2 - DOCUMENTATION

Sift: a local verification layer for AI coding agents

Individual technical report: performance, testing, explainability, ablation, data and model cards

Course	23CSE473 Neural Networks and Deep Learning
Professor	Dr. T. Senthil Kumar, Professor, CSE Department
Team	Team / Group 24 (single-member team)

SUBMITTED BY

Kundhave S

CB.SC.U4CSE23037

Scope: single-member team; all design, implementation, experiments, testing and documentation were done by Kundhave S.

Review 2 - Individual Documentation

Following the headings of the CDAO Test and Evaluation of AI Models framework (April 2024)

PROJECT	DELIVERABLE	TEAM NO	TEAM MEMBER	TEST COHORT
Sift	**Review 2 Documentation**	**Team 24 (single member)**	**Kundhave S, CB.SC.U4CSE23037**	**2,700 hunks (Python, JavaScript)**

1. Project overview and individual scope

Sift reads the intent a developer expressed over a coding-agent session and classifies every diff hunk as Aligned or Misaligned with it, then ranks hunks so a reviewer sees the likeliest misaligned ones first. Misaligned hunks carry one of three drift types: Collateral, Scope-Expansion, Purpose-Drift. Everything runs locally. The model is a ModernBERT-base cross-encoder over (consolidated intent, hunk) with alignment, drift-type and confidence heads, plus a distilled MiniLM student exported to INT8 ONNX for CPU use.

Novelty is not claimed for any component. The encoder and the training methods (focal loss, multi-task learning, temperature scaling, distillation, confident learning) are existing. The contribution is the task framing (hunk-level, multi-turn consolidated intent), the pipeline combining them, and the Sift-IA v1.0 dataset. **Individual scope:** single-member team, so all work (dataset use, model, experiments, tests, UI, reports) is by Kundhave S.

2. Performance

Test split, 2,700 hunks, labels *label_clean*, decision threshold tuned on validation. Pairwise AUC is within-session, per difficulty tier. NDCG@5 is over rankable sessions (at least 2 hunks, at least 1 misaligned).

Model	macro-F1	Prec.	Recall	NDCG@5	AUC random	AUC hard	ECE
B2 Diff lines changed	-	-	-	0.724	-	-	-
B4 TF-IDF dissimilarity	-	-	-	0.830	0.812	0.551	-
B5 Logistic regression	0.698	0.506	0.574	0.816	0.777	0.551	-
Siamese BiLSTM	0.562	0.309	0.362	0.783	0.641	0.526	0.234
MiniLM-L6 cross-encoder	0.616	0.410	0.382	0.835	0.741	0.636	0.059
CodeBERT cross-encoder	0.460	0.246	0.667	0.779	0.549	0.600	0.103
Frozen MiniLM cosine	0.614	-	-	0.874	0.814	0.719	-
GAN-style (hybrid)	0.625	-	-	0.871	0.807	0.705	-
Main: ModernBERT-base	0.680	0.516	0.476	0.868	0.800	0.726	0.066
Distilled MiniLM student (fp32)	0.667	0.521	0.423	0.869	0.794	0.678	0.107
Student ONNX INT8 (deployed target)	0.667	0.497	0.453	0.869	-	-	0.051

Table 1: headline results (results/master_comparison.csv). A dash means the baseline does not produce that quantity.

The main model has the highest hard-tier AUC (0.726), but by a small margin over frozen cosine dissimilarity (0.719); the frozen-embedding baseline has a slightly higher NDCG@5 (0.874 vs 0.868). On macro-F1 the main model (0.680) is below the logistic-regression baseline B5 (0.698). NDCG@5 has a high floor (0.7236 for random order) because the median session has 2 hunks.

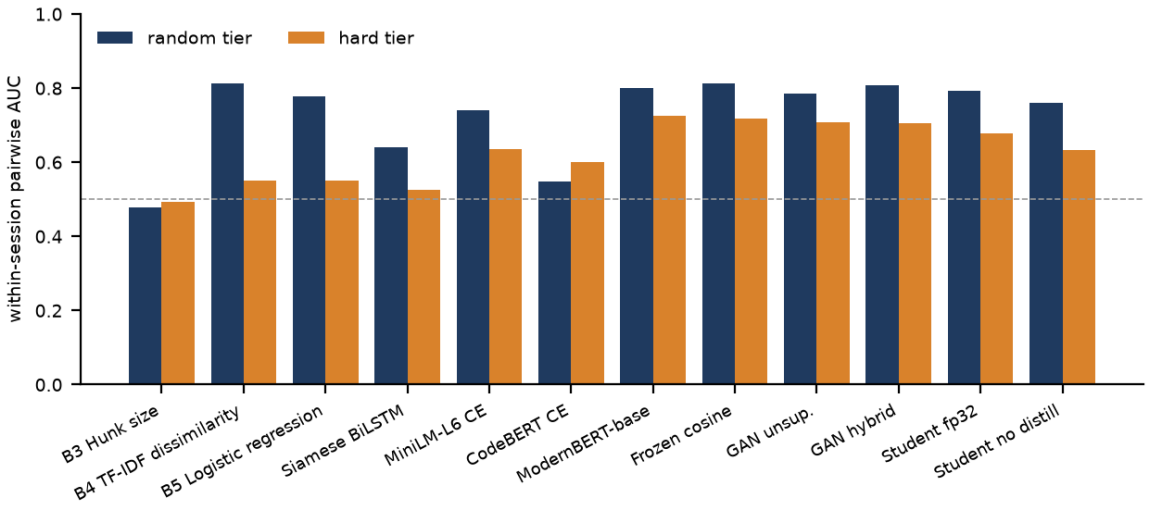


Figure 1: within-session pairwise AUC per tier (results/master_comparison.csv; models without stored tier AUC omitted).

Latency and size (measured on this machine only)

Model / runtime	Size MB	p50 ms	p95 ms	Conditions
ModernBERT-base, torch fp32, CPU	596.1	94.5	95.1	13th Gen Intel(R) Core(TM) i7-13650HX, 14 threads, L=512, batch 1, idle
ModernBERT-base, torch fp32, GPU	596.1	11.6	11.9	NVIDIA GeForce RTX 4050 Laptop GPU
MiniLM student, ONNX fp32, CPU	90.4	13.4	68.7	4 threads; CPU shared with a GPU training job
MiniLM student, ONNX INT8, CPU	22.9	8.5	66.9	same; p95 may be pessimistic

Table 2: size and latency (results/latency_full.json, results/latency_size_onnx.json). The INT8 student is 22.9 MB, meeting the planned under-30 MB target; the planned under-100 ms p95 on CPU is met by both the full model (95.1 ms) and the INT8 student, but the INT8 p95 was measured while the CPU was shared, so it is not a clean figure. INT8 vs fp32 logit correlation: 0.969.

Other dimensions

- **Interpretability:** Integrated Gradients (24 steps, PAD baseline). Masking the top-10 attributed tokens lowers P(misaligned) by 0.295 on average versus 0.173 for 10 random tokens (100 test hunks predicted misaligned); attribution beats random in 64% of hunks. Input x Gradient gives a larger gap (0.351 vs 0.146).
- **Robustness:** red-team defects D3 and D4 are open. Comment line copying the intent text added to a misaligned hunk flips 68% (17/25) of detected misaligned hunks to aligned; the 'this change implements' context comment flips 52% (13/25). The model partly keys on lexical overlap with the intent.
- **Uncertainty and calibration:** temperature scaling (T = 1.275, fitted on validation) reduces test ECE from 0.064 to 0.028. The table above reports ECE of the raw evaluation runs.
- **Drift:** see Scope and limitations. Proposed procedure: compute the model's score distribution and the share of flagged hunks on new repositories or a later time window, compare with the test distribution with a two-sample test, and re-evaluate on freshly labelled samples. Commit dates reach 2026, but the splits are by repository, not by time.
- **Representativeness:** splits are repository-disjoint. Test repositories: attrs, axios, httpx (1,686 Python and 1,014 JavaScript hunks). **The test split has no TypeScript** (all TypeScript is in validation), so TypeScript generalisation is untested.
- **Bias:** The only imbalances identified are language and repository (Python, JavaScript and TypeScript only, 15 open-source projects). No analysis of developer demographics or other fairness dimensions was done.

3. Testing methods

The pytest suite has 220 tests: 214 passed, 6 expected failures (strict xfail, each a recorded defect), 0 failed, 0 skipped. Line coverage of src/sift and app/backend is 73.4% (src/sift + app/backend (unit/API tests; e2e not measured)). Categories cover smoke, white-box, black-box, data pipeline (including zero repositories shared across splits), metamorphic relations, adversarial/red-team, API, back-to-back (full model vs student) and browser end-to-end tests. Details are in the separate Testing document.

Category	Files	Passed	xfailed	Failed	Framework section
adversarial	test_redteam.py	17	2	0	Testing Methods

Category	Files	Passed	xfailed	Failed	Framework section
api	test_api.py, test_api_partitions.py	43	0	0	Testing Methods
back-to-back	test_backtoback.py	4	0	0	AI Models
black-box	test_blackbox.py	31	2	0	Testing Methods
data-pipeline	test_data_pipeline.py	17	2	0	Data
e2e	test_ui_e2e.py	37	0	0	Context
metamorphic	test_metamorphic.py	8	0	0	Testing Methods
smoke	test_smoke.py	6	0	0	Performance / readiness
white-box	test_diffparse_consolidate.py, test_whitebox.py	51	0	0	AI Models

Table 3: test results by category (results/test_summary.json).

4. Explainability (XAI)

Token-level attribution on the cross-encoder uses captum Integrated Gradients (target: P(misaligned), baseline: [PAD] embedding at all non-special positions, 24 steps) and Input x Gradient, aggregated to words and diff lines. On average 19.0% of absolute attribution falls on intent tokens. Faithfulness uses replace-by-[PAD] deletion on 100 test hunks predicted misaligned:

Top-k masked	IG drop (top)	IG drop (random)	IG flip rate (top / random)	IxG drop (top)	IxG drop (random)
3	0.084	0.058	27% / 20%	0.150	0.045
5	0.192	0.097	46% / 27%	0.243	0.105
10	0.295	0.173	58% / 44%	0.351	0.146
20	0.380	0.279	74% / 57%	0.402	0.279

Table 4: faithfulness (results/xai.json). Explanation costs a median 506 ms on the NVIDIA GeForce RTX 4050 Laptop GPU versus 25 ms to score (may be inflated if other jobs shared the GPU). Attribution is evidence about what the model used, not proof of the cause.

5. Ablation

All variants use the same protocol and test split, one seed (473). Differences of about 0.01 macro-F1 are within expected run-to-run noise and are not significant.

Configuration	macro-F1	NDCG@5	AUC rand.	AUC hard	ECE	Size MB
1. Full model	0.680	0.868	0.800	0.726	0.066	596.1
2. hunk-only	0.462	0.780	0.561	0.592	0.110	596.1
3. intent-only	0.547	0.726	0.505	0.501	0.109	596.1
4. no intent consolidation	0.689	0.891	0.831	0.738	0.057	596.1
5. no focal loss (CE)	0.696	0.887	0.844	0.712	0.145	596.1
6. alignment head only	0.706	0.895	0.857	0.752	0.082	596.1
7. no denoising	0.688	0.902	0.861	0.767	0.119	596.1
10. Distilled MiniLM student (fp32)	0.667	0.869	0.794	0.678	0.107	90.9
11. INT8 ONNX student (deployed target)	0.667	0.869	-	-	0.051	22.9

Table 5: ablation (results/ablation.csv; INT8 per-tier AUC was not stored).

Findings: removing the hunk or the intent clearly hurts (hunk-only and intent-only are near chance on the hard tier), which supports reading both texts jointly. In contrast, the ablated variants without consolidation, focal loss, multi-task heads or denoising did **not** underperform the full model on macro-F1 in this single-seed run, so these components are not shown to help here.

6. Baseline comparison

Table 1 and Figure 1 give the comparison: five non-neural baselines (B1-B5), a Siamese BiLSTM, a MiniLM-L6 cross-encoder and a reduced-protocol CodeBERT cross-encoder, frozen-embedding cosine dissimilarity, and two GAN-style adversarially trained baselines on frozen MiniLM embeddings (an interpretation of the GAN deliverable, since Sift has no GAN). The best non-neural hard-tier AUC is 0.551 (B4); the main model reaches 0.726. A frozen-embedding cosine baseline needs no training and is close to the main model on ranking (0.719 hard AUC), while its macro-F1 is lower (0.614). The GAN-style baselines did not beat the cross-encoder on hard-tier AUC (0.705 for the hybrid). UniXcoder was dropped, and the CodeBERT run used a reduced protocol.

7. Data and dataset

Data Card: Sift-IA v1.0

Field	Value
Name / version	Sift-IA v1.0 (released package; the builder scripts are not in this workspace)
Source	15 open-source GitHub repositories, licences: Apache-2.0 (requests); BSD-3-Clause (flask, click, httpx); MIT (rich, typer, attrs, black, sqlmodel, express, axios, got, chalk, fastify, koa). Licence texts are shipped with the package.
Size	18,000 hunk-intent pairs, 6,111 sessions, 8,016 distinct source commits; median 132 tokens per pair, p95 455, max 4,031
Splits (by repository)	train: 12,600 hunks, 4,266 sessions, 2,772 misaligned (black, click, express, fastify, flask, requests, rich); val: 2,700 hunks, 921 sessions, 594 misaligned (chalk, got, koa, sqlmodel, typer); test: 2,700 hunks, 924 sessions, 594 misaligned (attrs, axios, httpx)
Label construction	Aligned hunks are real hunks of the seed commit; a misaligned hunk is a donor hunk from a different commit inserted into the session (tiers: random, hard by identifier-overlap mining). Labels follow from construction. Drift types: Collateral, Scope-Expansion, Purpose-Drift.
Noise shipped with ground truth	4,529 rows carry noise: intent turns 2,730; duplicate hunks 937; non-semantic hunks 714; label flips 569 (train only); over-length 198
Collection dates	commits from 2009-06-26 to 2026-09-19
Known limitations	Misaligned hunks are constructed, so a model may learn construction artefacts (the hard tier mitigates, not removes this). No human labels (the 900-sample gold sheet is blank). Seed commits may be tangled. A donor hunk may coincidentally suit the intent. Only Python, JavaScript, TypeScript. Dataset audit: the released per-language hunk counts differ from the earlier documented counts (measured Python 10,205, JavaScript 6,726, TypeScript 1,069); the test split has no TypeScript; 456 rankable test sessions, not 447.
Intended use	Research and benchmarking of hunk-level intent-alignment classification and ranking.
Not for use	Judging real developers, defect prediction, security auditing, or any claim about agent-authored code (v1.0 has no agent-authored sessions).

Table 6: Data Card (data/dataset_statistics.json, data/source_repositories.csv, experiments/results/data_verification.json).

8. AI models

Model Card: Sift ModernBERT cross-encoder and INT8 student

Field	Value
Model details	answerdotai/ModernBERT-base cross-encoder, 149,018,117 parameters, three heads (alignment, 3-way drift type, confidence) on the pooled representation; input is consolidated intent plus hunk text, max length 512. Deployed variant: 22M-parameter MiniLM student distilled (tau 4.0, alpha 0.5) and exported to INT8 ONNX.
Intended use	Local triage: rank the hunks of one diff by misalignment probability so a human reviews the likeliest ones first, with calibrated confidence and token attributions.
Out-of-scope use	Automatic accept/reject of changes; security or correctness verification; languages other than Python/JavaScript/TypeScript; decisions about people.
Training data	Sift-IA v1.0 train split (12,600 rows); 10,128 used after rule filters (removed 384 non-semantic, 502 duplicate) and confident-learning removal (1,586). Validation used for threshold, early stopping and temperature.
Training settings	AdamW, lr 2e-05, micro-batch 8 x accumulation 2 (effective 16), up to 6 epochs with early stopping (patience 2) on validation macro-F1, focal loss, loss weights 0.6/0.3/0.1, bf16 autocast, gradient checkpointing, seed 473. Best validation macro-F1 0.716; training time 28 min on NVIDIA GeForce RTX 4050 Laptop GPU (peak 3.1 GB). Context length was reduced from the planned 1,024 to 512. Hyperparameter search: 6 TPE trials (planned 60) on a 1600-row subset, 2 epochs.
Metrics (test)	Main: macro-F1 0.680, precision 0.516, recall 0.476, specificity 0.874, NDCG@5 0.868, AUC random/hard 0.800/0.726. INT8 student: macro-F1 0.667, NDCG@5 0.869, 22.9 MB.
Calibration	Temperature scaling, T = 1.275; test ECE 0.064 to 0.028.
Limitations	Single seed. Constructed labels with no human validation. Open misleading-comment weakness (D3, D4). Recall on misaligned hunks is below 0.5. No TypeScript in the test split. Confident learning found only a fraction of the known label flips (precision 15%, recall 39% against the 569 known flips), and the reduced noise-robustness study (8 runs) shows no consistent benefit from denoising.
Ethical considerations	Built only on permissively licensed public code; no personal data was added. Risk: over-trust. A flagged or unflagged hunk is a ranking signal for a human reviewer, not a verdict, and false negatives are common.
How to run	`PYTHONPATH=src .venv/bin/python -m sift.infer --help` for the CLI; the web workspace and API are in app/. Checkpoints are in artifacts/.

Table 7: Model Card (experiments/runs/full/config.json and results.json, results/calibration.json, results/hpo_small.json, results/confident_learning.json, results/noise_robustness.json).

9. Context

Use case: a developer reviews an AI agent's diff and wants the hunks that do not serve the stated request first. **Environment:** everything runs locally; the deployed target is the INT8 student on CPU. Measurements were made on 13th Gen Intel(R) Core(TM) i7-13650HX (20 logical cores) and a NVIDIA GeForce RTX 4050 Laptop GPU used for training; no other hardware was tested, so the latency figures should not be generalised. No network service or database is required.

10. Defect register summary

ID	Severity	Description	Status
D-DATA-1	Low	documented counts are 11,900/5,076/1,024 hunks per language; released data differs	open (xfail strict)
D-DATA-2	Medium	test split has no TypeScript although val does; TS generalisation is untested	open (xfail strict)
D1	Medium	plain `diff -u` multi-file output (no `diff --git` lines) merges the second file's header into the previous hunk	open (xfail strict)
D2	Low	chit-chat filter is exact-match on 8 template phrases; paraphrased pleasantries are kept in the intent	open (xfail strict)
D3	High	comment line copying the intent text added to a misaligned hunk flips 68% (17/25) of detected hunks to 'aligned'; threshold 20%	open (xfail strict)
D4	Medium	context comment 'this change implements: <intent>' flips 52% (13/25) of detected hunks to 'aligned'; threshold 20%	open (xfail strict)

Table 8: all 6 defects are open and recorded as strict xfail tests (results/test_summary.json). Mitigations are given in the Testing document.

11. Scope and limitations

Scope notes that apply to the reported numbers: one seed (473) per run; labels are construction-determined with no human validation; the test set is repository-disjoint but is open-source Python/JavaScript only; latency and size are valid only for the machine measured; noise-robustness and baselines used reduced settings; the hyperparameter search ran 6 trials, not the planned 60; the UniXcoder comparison is outside this work; drift and bias were not measured, and the rules-only denoising ablation is outside this work; the GAN-style baseline is our interpretation of the GAN comparison requirement.

12. Conclusion

- The ModernBERT cross-encoder has the best hard-tier pairwise AUC (0.726) and clearly beats the non-neural baselines there (best 0.551). It does **not** win on macro-F1, where B5 and several ablated variants score higher, and a frozen-embedding cosine baseline matches it closely on ranking.
- The deployed INT8 student is 22.9 MB and keeps macro-F1 0.667 and NDCG@5 0.869, close to the full model, at a CPU p50 of 8.5 ms (shared-CPU measurement).
- Individual components (consolidation, focal loss, multi-task heads, denoising) are not shown to help in this single-seed study; only the use of both intent and hunk is clearly supported.
- Not done: multiple seeds, human-validated labels, the full 60-trial hyperparameter search, UniXcoder, full-protocol noise robustness, drift and bias measurement, agent-authored sessions, TypeScript test evidence, and fixes for open defects D1-D4.
- Novelty: none of the methods is new; the contribution is the task framing, the pipeline, and the dataset.